

Statistics

Worksheet 1b

Mean, median and mode of grouped data (Class Interval)

1. S21/qp21/q9(b)

The times taken by the 80 students to complete a Science test are shown in the frequency table.

Time (m minutes)	$40 < m \leq 50$	$50 < m \leq 60$	$60 < m \leq 70$	$70 < m \leq 80$	$80 < m \leq 90$
Frequency	8	13	p	20	q

An estimate for the mean time taken to complete the test is 67.625 minutes. This is calculated using the mid-interval value as an estimate of the time in each interval.

Calculate the value of p and the value of q . [5]

2. S20/qp21/q1(a)

The speeds, v km/h, of 80 vehicles travelling along a road were recorded. The results are shown in the table.

Speed (v km/h)	Frequency
$30 < v \leq 40$	10
$40 < v \leq 50$	18
$50 < v \leq 60$	27
$60 < v \leq 70$	19
$70 < v \leq 80$	6

Calculate an estimate of the mean speed of the vehicles. [3]

3. S19/qp22/q8(b)

The mass, in grams, of each of 75 oranges is measured. The results are summarised in the table.

Mass (m g)	Frequency
$160 < m \leq 170$	4
$170 < m \leq 180$	10
$180 < m \leq 190$	15
$190 < m \leq 200$	23
$200 < m \leq 210$	18
$210 < m \leq 220$	5

Calculate an estimate of the mean mass.

[3]

4. S19/qp21/q8(b)

The table summarises the distances, dm, that 80 women threw the javelin.

Distance (d m)	Frequency
$20 < d \leq 25$	6
$25 < d \leq 30$	16
$30 < d \leq 35$	25
$35 < d \leq 40$	18
$40 < d \leq 45$	13
$45 < d \leq 50$	2

Calculate an estimate of the mean distance the javelin was thrown.

[3]

5. W18/qp21/q1(b)(i)(ii)

The table summarises the times taken by the 120 employees in a company to travel to work.

Time (t minutes)	$0 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 60$	$60 < t \leq 80$	$80 < t \leq 100$
Frequency	29	38	26	21	6

- Write down the modal class. [1]
- Calculate an estimate of the mean time. [3]

6. S18/qp22/q2(b)

Jenny asked 60 people how many movies they had each watched in the last month. The table summarises her results.

Number of movies	0	1	2	3	4	5	6
Frequency	p	14	15	7	q	5	2

The mean number of movies watched is 2.3.

Find the value of p and the value of q . [3]

7. W17/qp21/q2(a)

Sunil recorded the lengths, in minutes, of the 150 phone calls he made one month. His results are summarised in the table.

Length of call (t minutes)	$0 < t \leq 5$	$5 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 50$
Frequency	35	42	30	28	15

Calculate an estimate of the mean length of a call. [3]

8. S17/qp22/q11(a)(i)(ii)

The table below summarises the times taken by 50 athletes to run 400m.

Time (t seconds)	$50 \leq t < 55$	$55 \leq t < 60$	$60 \leq t < 65$	$65 \leq t < 70$	$70 \leq t < 75$
Frequency	7	16	15	11	1

- State the modal class. [1]
- Calculate an estimate of the mean time taken by these athletes. [3]

9. W16/qp21/q10(a)

The times taken by 135 runners to complete a cross-country course were recorded.
The results are summarised in the table.

Time (t minutes)	$20 < t \leq 30$	$30 < t \leq 35$	$35 < t \leq 40$	$40 < t \leq 50$	$50 < t \leq 70$
Number of runners	15	30	40	35	15

- Calculate an estimate of the mean time. [3]

Mathlete
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Answer Key

1. S21/qp21/q9(b)

9(b)	$p = 24$ $q = 15$	5	B1 for $p + q = 39$ soi B1 for correct midpoints soi M1 for $\frac{8 \times 45 + 13 \times 55 + 65p + 20 \times 75 + 85q}{80}$ M1 for correct elimination of p or q
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2. S20/qp21/q1(a)

1(a)	54.1 or 54.12 to 54.13 nfw	3	B1 for use of correct midpoints M1 for $\frac{35 \times 10 + 45 \times 18 + 55 \times 27 + 65 \times 19 + 75 \times 6}{80}$
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3. S19/qp22/q8(b)

8(b)	192 or 192.4 to 192.5 nfw	3	M1 for $4 \times 165, 10 \times 175, 15 \times 185, 23 \times 195,$ 18×205 and 5×215 M1 for $(4 \times 165 + 10 \times 175 + 15 \times 185 + 23 \times 195 +$ $18 \times 205 + 5 \times 215) \div 75$ oe After 0 scored, SC1 for $\sum fx \div 75$
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4. S19/qp21/q8(b)

8(b)	33.875 or 33.9 nfw	3	B1 for correct use of midpoints M1 for $\frac{\sum fx}{80}$
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5. W18/qp21/q1(b)(i)(ii)

1(b)(i)	$20 < t \leq 40$	1	
1(b)(ii)	39.5	3	B1 for correct use of midpoints soi M1 for $(10 \times 29 + 30 \times 38 + 50 \times 26 + 70 \times 21 + 90$ $\times 6) \div 120$ oe

6. S18/qp22/q2(b)

2(b)	$q = 9$	B2	M1 for $[0 \times p] + 1 \times 14 + 2 \times 15 + 3 \times 7 + 4 \times q + 5 \times 5 + 6 \times 2$ oe
	$p = 17 - \text{their } q$	B1	Strict FT provided q integer with $0 \leq q \leq 17$

7. W17/qp21/q2(a)

2(a)	14.35 or 14.4	3	B1 for use of correct midpoints soi M1 for $(2.5 \times 35 + 7.5 \times 42 + 15 \times 30 + 25 \times 28 + 40 \times 15) \div 150$
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8. S17/qp22/q11(a)(i)(ii)

11(a)(i)	$55 \leq t < 60$	1	
11(a)(ii)	60.8 nfw	3	M2 for $\frac{\sum \text{frequency} \times \text{midvalue}}{50}$ oe or M1 for $\sum ft$

9. W16/qp21/q10(a)

10 (a) (i)	Correct histogram with linear scale on frequency density axis	3	B2 for all 5 heights correct with axis scaled OR B1 for at least 3 correct frequency densities soi and B1 for all 5 bars correct widths
(ii)	39.4[4...]	3	B1 for use of correct midpoints M1 for $\frac{\sum fx}{135}$

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